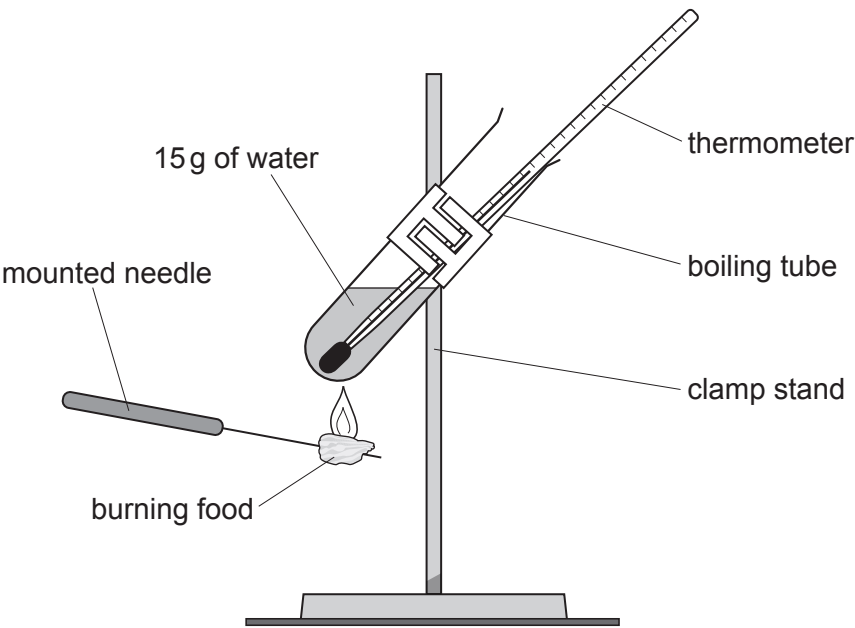


4. Students were investigating the energy content of food. They heated 15g of water with each sample of food.



Their results are shown in the table below.

Food	Mass of food burnt (g)	Initial water temperature (°C)	Final water temperature (°C)	Rise in temperature of water (°C)
bread	2.5	18	23	5
cheese	4.0		31	14
cornflakes	3.0	17	29	12
oatie biscuit	5.0	18	25	7

(a) Complete the table. [1]

(b) Use the equation:
energy released (J) = mass of water (g) × 4.2 × temperature increase (°C)
to calculate the energy released by the cheese. [2]

energy released = J



- (c) Use the students' results to calculate the energy provided by a snack consisting of 20 g of cheese and 15 g of oatie biscuits. [3]

energy = J

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